

Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

TEAM ID	SWTID-2026-2268
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Step 1: Monitor Data Rendered to Database

- Open MySQL Workbench.
- Select the electricity database.
- Expand the tables and choose the required table.
- Click the (i) Information icon.
- Check details such as table rows, column count, and storage information.
- Verify that only the required amount of data is being retrieved to improve performance.

Step 2: Optimize Data Retrieval

- Ensure database queries load only necessary records.
- Reduce unnecessary data processing to avoid system overload.
- Improve dashboard response time and efficiency.

Step 3: Apply Data Filters in Tableau

- Create filters to limit the data displayed.
- Use filters on State, Region, Year, and Date.
- Enable users to analyze specific segments of electricity consumption data.

Step 4: Use Top N and Bottom N Filters

- Identify the highest and lowest electricity-consuming states.

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- Reduce the amount of data displayed and improve visualization performance.

Step 5: Verify Dashboard Performance

- Check whether visualizations load quickly.
- Ensure interactive filters respond efficiently.
- Confirm that the dashboard provides accurate and meaningful insights.

Step 6: Test Visualizations

Validate the performance of the following charts:

- 2019 State Consumption
- 2020 State Consumption
- Total Consumption
- Usage by Region
- Top N and Bottom N Analysis
- Month-wise Consumption (2019 & 2020)
- Region-wise Total Consumption
- Usage Before and After Lockdown
- Region-wise State Usage
- Quarter-wise Usage
- Metro City State Usage
- Usage by Year

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